Connor Brooks

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Education

University of Iowa, Iowa City, IA — August 2025 - May 2027
B.S. in Mathematics (Computer Science track) — GPA: 4.10

Experience

Math and Physics Tutor — University of Iowa — September 2026 – Present
Student-Athlete Academic Services — Iowa City, IA

- Tutored student-athletes in Algebra, Precalculus, Calculus, Vector Calculus, and Physics classes
- Developed individualized lesson plans for each student ahead of weekly hour-long tutoring sessions

Research Assistant — Robotic Motion Planning — University of Iowa — August 2026 – Present
Advisor: Dr. Shaoping Xiao — Iowa City, IA

- Researched Reinforcement Learning in Partially Observable environments for robotic motion planning
- Employed a Deep Recurrent Q Network to handle a sequence of imperfect observations as input
- Conducted literature review of Reinforcement Learning for Robotic Motion Planning with LTL constraints

Research Assistant — Ag Robotics — Oregon State University Robotics REU — June 2026 – August 2026
Advisor: Dr. Cindy Grimm — Corvallis, OR

- Created a data pipeline for grounding an apple tree simulation to observations from real-world data
- Wrote an Impedance Controller for manipulating the apple tree with a Franka Research 3 arm
- Used OpenCV to track positions of parts of the apple tree during the arm's manipulation

Backend Engineer — Aug 2025 – Oct 2025
Contract — Remote

- Built the FastAPI backend of a web-app platform for Christmas-light installation companies to generate marketing materials and track marketing campaigns
- Integrated Stripe API for handling subscription billing

Projects

Arduino Mini Projects *(C, CAD, 3D Printing)* — April 2026 – May 2026

- Self-studied electronics with an Arduino board via an assortment of mini projects
- Programmed, Wired, and Designed an alarm clock as the final project

Chess AI *(Python)* — Feb 2025 – May 2025

- Studied classical AI by implementing minimax search, alpha-beta pruning, and evaluation heuristics for chess

Technical Skills

Languages: Python, Java, C, SQL
Tools: PyTorch, NumPy, OpenCV, libfranka, ROS2, FastAPI
Other: LaTeX, Git, GitHub

Volunteer Work

Vice President — Iowa Math Club — May 2026 – Present
Mentor — FIRST Tech Challenge (FTC) High School Team — Sept 2025 – Present
Volunteer — Iowa City Homeless Project — Aug 2025 – Present

Awards

Undergraduate Research Assistantship Award — UIowa Math Dept. — Sept 2026
Principal Scholar Award — Principal Financial Group — July 2026
Richard J. Odem and Robin W. Odem Scholarship — UIowa Math Dept. — March 2026
National Merit Scholar — National Merit Scholarship Corporation — May 2025
Joann M. Stuhr Scholarship — May 2025
Iowa Governor's Scholar — April 2025
Iowa Flagship Scholarship — University of Iowa — March 2025